

Worksheet No. - 1

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Experiment 1: Python Libraries and Data Preprocessing on Netflix Dataset. a. Math b. NumPy c. Matplotlib d. Seaborn e. SciPy.

ii) Perform comprehensive data preprocessing on the **Netflix dataset** by handling missing values, detecting and treating outliers, encoding categorical features, applying feature scaling, and removing noise/duplicate records to improve overall data quality and prepare the dataset for machine learning tasks.

1.Aim:

To understand the usage of fundamental Python libraries for Machine Learning and perform comprehensive data preprocessing on the Netflix dataset.

2.Objective:

1. To load and explore the Netflix dataset using Pandas.
2. To identify missing values in the dataset.
3. To handle missing values using appropriate techniques.
4. To convert categorical data into numerical format using Label Encoding.
5. To normalize numerical data using Min-Max Scaling.
6. To standardize numerical data using StandardScaler.

3.Task to be done :

- Import the required Python libraries (Pandas and Scikit-learn).
- Load the Netflix dataset from the CSV file.
- Display the first five rows of the dataset.
- Examine the dataset structure and identify missing values.
- Fill missing values in categorical columns with "Unknown".
- Fill missing values in numerical columns using the mean of each column.
- Encode categorical features using LabelEncoder.

- Normalize numerical features using MinMaxScaler.
- Standardize numerical features using StandardScaler.

4. Online Resources:

1. NumPy Documentation
2. Pandas Documentation
3. Scikit-Learn Official Documentation
4. Matplotlib Official Documentation
5. Seaborn Documentation
6. Kaggle Titanic Dataset Resources

Software Requirements

1. Python 3.x
2. Jupyter Notebook / Google Colab
3. Anaconda Distribution

Standard Tools Used

1. NumPy
2. Pandas
3. Matplotlib
4. Seaborn
5. SciPy
6. Scikit-Learn

5. Procedure/ Algorithm:

```
import pandas as pd
from sklearn.preprocessing import LabelEncoder, StandardScaler

# Load the dataset
df = pd.read_csv(r"C:\Users\ASUS\Downloads\archive\netflix.csv")

# =====
# 1. Find Missing Values
# =====
print("Missing Values Before Filling:\n")
print(df.isnull().sum())

# =====
# 2. Fill Missing Values
# =====

# Separate categorical and numerical columns
categorical_cols = df.select_dtypes(include='object').columns
numerical_cols = df.select_dtypes(include=['int64', 'float64']).columns

# Fill categorical columns with "Unknown"
for col in categorical_cols:
    df[col] = df[col].fillna("Unknown")
```

```
# Fill numerical columns with mean
for col in numerical_cols:
    df[col] = df[col].fillna(df[col].mean)
print("\nMissing Values After Filling:\n")
print(df.isnull().sum())
```

3. Label Encoding

```
encoder = LabelEncoder()

for col in categorical_cols:
    df[col] = encoder.fit_transform(df[col])

print("\nFirst 5 Rows After Encoding:\n")
print(df.head())
```

5. Standardize Numerical Values

```
standard_scaler = StandardScaler()

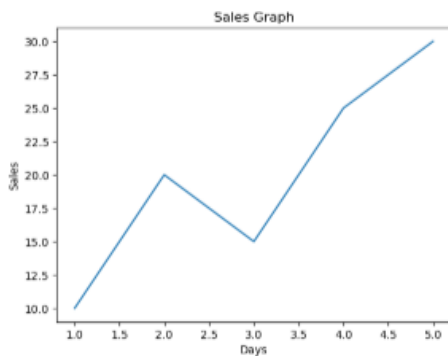
if len(numerical_cols) > 0:
    df[numerical_cols] = standard_scaler.fit_transform(df[numerical_cols])

print("\nStandardized Numerical Columns:\n")
print(df[numerical_cols].head())
```

Final Dataset

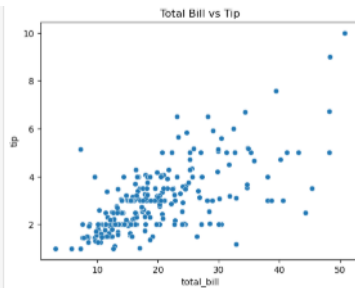
```
print("\nFinal Dataset Preview:\n")
print(df.head()) .....continue,
```

6.output:



```
Array: [10 20 30 40 50]
Mean: 30.0
Sum: 150
Maximum: 50
Minimum: 10
```

```
Square Root: 5.0
Power: 8.0
Factorial: 120
Value of Pi: 3.141592653589793
Cos 0: 1.0
```



```
===== DATASET SHAPE =====
(7787, 12)
```

```
===== DATASET INFORMATION =====
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7787 entries, 0 to 7786
Data columns (total 12 columns):
#   Column          Non-Null Count  Dtype
---  -
0   show_id         7787 non-null   object
1   type            7787 non-null   object
2   title           7787 non-null   object
3   director        5398 non-null   object
4   cast            7069 non-null   object
5   country         7280 non-null   object
6   date_added      7777 non-null   object
7   release_year    7787 non-null   int64
8   rating          7780 non-null   object
9   duration        7787 non-null   int64
10  genres          7787 non-null   object
11  description     7787 non-null   object
dtypes: int64(2), object(10)
memory usage: 730.2+ KB
None
```

```
===== MISSING VALUES BEFORE FILLING =====
```

```
show_id      0
type         0
title        0
director     2389
cast         718
country      507
date_added   10
release_year  0
rating       7
duration     0
genres       0
description  0
dtype: int64
```

===== MISSING VALUES AFTER FILLING =====

```
show_id      0
type         0
title        0
director     0
cast         0
country      0
date_added   0
release_year 0
rating       0
duration     0
genres       0
description   0
dtype: int64
```

Duplicate Rows : 0

Dataset Shape After Removing Duplicates : (7787, 12)

===== DATA AFTER LABEL ENCODING =====

	show_id	type	title	director	cast	country	date_added	release_year \
0	0	1	83	3846	3160	39	328	2020
1	1	0	47	3892	5160	229	385	2008
2	2	0	79	1473	5250	249	1358	2016
3	3	0	989	2159	267	549	110	2016
4	4	1	990	3846	1513	462	1177	2017

	rating	duration	genres	description
0	8	4	398	3912
1	8	143	339	1973
2	9	124	315	6356
3	5	90	177	4791
4	7	1	95	6304

===== STANDARDIZED NUMERICAL DATA =====

	release_year	duration
0	0.692878	-1.278234
1	-0.677480	1.450067
2	0.236092	1.077134
3	0.236092	0.409780
4	0.350289	-1.337118

6.Learning outcomes (what I have learnt) :

After completing this experiment, students will be able to:

- Understand the importance of data preprocessing in machine learning.
- Load and explore a dataset using the Pandas library.
- Handle missing values and remove duplicate records.
- Apply Label Encoding to convert categorical data into numerical form.
- Standardize numerical data using StandardScaler.
- Split the dataset into training and testing sets.
- Visualize data using bar and pie charts with Matplotlib.
- Prepare a clean dataset for machine learning and data analysis.